DT 111 · TOPIC 5

Introduction to Machine Learning

Supervised and unsupervised learning, model training and evaluation

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BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain where Machine Learning fits within AI.
- Distinguish features, targets and common ML task types.
- Describe training, validation, testing and generalisation.
- Explain decision trees and K-means at an introductory level.
- Interpret accuracy, precision, recall and simple regression error.

Programming with explicit rules and learning from examples

Traditional rule-based program

- Developer writes the rules
- Data + rules → output
- Works well when rules are explicit and stable

Machine Learning

- Algorithm learns patterns from examples
- Training data → model
- Useful when rules are difficult to specify directly

Features and target

Features

Input variables used by the model, such as attendance, coursework and study hours.

Target

The outcome the model should predict, such as final score or pass/fail.

Example row

Attendance 92%, coursework 78, study hours 8 → final score 82.

Supervised and unsupervised learning

Supervised learning

Training examples include known targets. Used for regression and classification.

Unsupervised learning

No target labels are supplied. The algorithm searches for structure such as clusters.

Three common Machine Learning tasks

Regression

Predict a number. Example: final score.

Classification

Predict a category. Example: Pass/Fail.

Clustering

Discover groups without predefined labels. Example: study profiles.

One dataset, different ML tasks

Attendance	Coursework	Study hours	Final score	Pass label
92	78	8	82	Pass
55	45	3	48	Fail
80	72	6	76	Pass
61	58	4	59	Pass

Regression target: final score.

Classification target: pass label.

Clustering can use the feature columns without a target.

What happens during training

1

Examples

Input features and targets

2

Model

Makes initial predictions

3

Error

Compare predictions with truth

4

Update

Adjust parameters or structure

5

Repeat

Reduce useful error patterns

Train, validation and test data

Keeping evaluation data separate provides a more honest estimate of generalisation.

1

Training set

Used to learn the model

2

Validation set

Used to tune choices and compare versions

3

Test set

Held back for final evaluation on unseen data

Decision trees

A decision tree predicts by following a sequence of feature-based questions.

Example: a simple decision tree

Real trees learn split rules from data; this is a simplified teaching example.

1

Attendance >= 70%?

No → Fail risk

2

Coursework >= 60?

Yes → continue

3

Study hours >= 4?

Yes → Pass

Why decision trees are intuitive

Readable logic

The path can often be explained as a sequence of conditions.

Mixed variables

Can work with several types of predictors after appropriate preparation.

Limitation

A deep tree can memorise training details and overfit.

K-means clustering

K-means groups observations by repeatedly assigning them to nearby centres and updating those centres.

K-means in four steps

1

Choose K

Number of clusters

2

Initial centres

Start with K centres

3

Assign

Place each point with nearest centre

4

Update

Move centres to cluster means

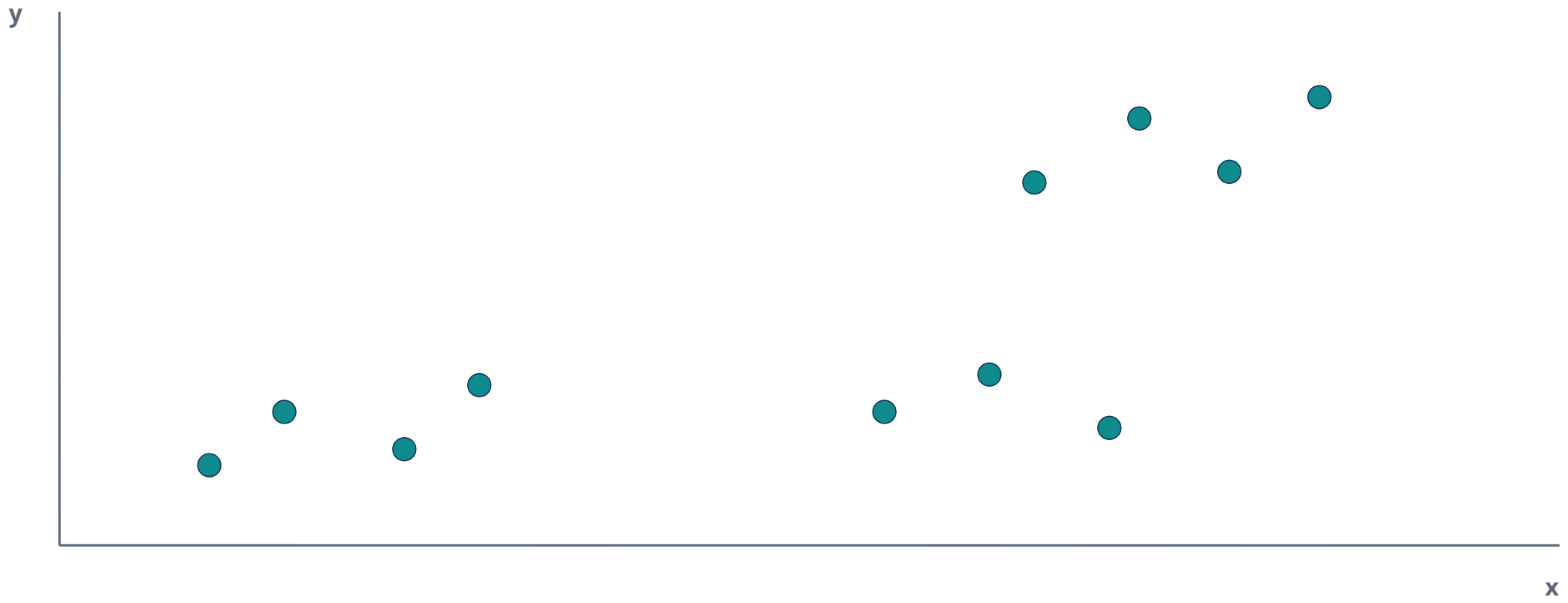
5

Repeat

Continue until stable

Example clustering pattern

Clusters can reveal structure, but people must decide whether the groups are meaningful in the real context.



Generalisation, underfitting and overfitting

Underfitting

Model is too simple to capture important structure.

Useful fit

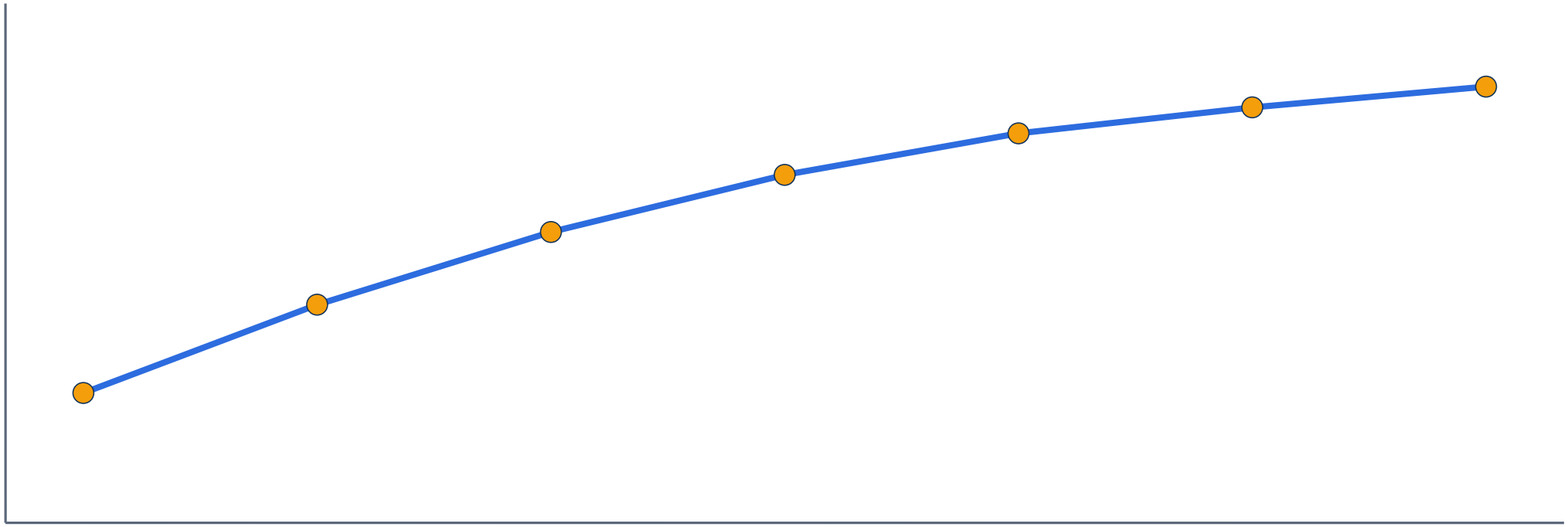
Captures stable patterns that also work on new data.

Overfitting

Learns training details too closely and performs poorly on unseen data.

Training and validation performance

Training performance may keep improving even after validation performance stops improving.





Evaluation depends on the task

A model is not “good” in the abstract. The evaluation metric should match the prediction problem and the consequences of errors.

Confusion matrix for classification

Classification errors are not all the same; the matrix separates true and false positive/negative outcomes.

		Predicted: Positive	Predicted: Negative
Actual	**Positive**	True Positive	False Negative
	Negative	False Positive	True Negative

Accuracy counts all correct predictions. Precision and recall focus on different kinds of positive prediction errors.

Precision and recall

Precision

- Of predicted positives, how many were truly positive?
- Important when false positives are costly

Recall

- Of true positives, how many did we find?
- Important when missing a positive case is costly

Regression evaluation

MAE

Average absolute difference between predicted and true values. Easy to interpret in the original units.

Residuals

Individual prediction errors can reveal systematic problems.

Context

An acceptable error depends on the application and decision consequences.

Model performance is not the whole project

Technical questions

- Does the model generalise?
- Are metrics appropriate?
- Is performance stable?

Human questions

- Is the target meaningful?
- Are the data representative?
- Are errors acceptable and fair?
- How will outputs be used?

Topic 5 key ideas

- Machine Learning learns patterns from data rather than relying only on explicitly written rules.
- Features are inputs; targets are outcomes to be predicted.
- Train, validation and test data serve different purposes.
- Overfitting is strong training performance with weak generalisation.
- Evaluation metrics must match the task and the consequences of errors.